

BROKEN HILL, AustraliaWMO: **946910**Lat: **32.0011S**Lon: **141.4694E**Elev: **924**StdP: **14.21**Time Zone: **9.50 (AUA)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	35.7	37.9	16.4	13.1	72.3	19.7	15.3	72.6	26.1	57.5	23.8	56.4	4.3	210	0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.2	104.0	65.8	100.5	65.1	97.2	64.4	72.3	84.7	70.3	85.4	68.5	86.4	13.2	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.3	111.4	76.1	66.4	100.5	74.9	63.2	89.6	75.0	36.7	83.6	34.8	85.5	33.2	85.7	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.4	22.9	20.7	31.6	107.8	1.6	3.3	30.4	110.1	29.4	112.1	28.5	114.0	27.3	116.4		
(5)			30.0	74.9	1.8	2.4	28.7	76.6	27.7	78.0	26.7	79.3	25.4	81.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	64.8	80.0	78.3	71.8	64.2	56.5	51.3	50.5	53.1	59.5	65.6	71.3	76.1	(6)
(7)		DBStd	12.42	8.37	8.05	7.83	6.80	5.63	4.83	4.52	5.48	7.00	7.76	8.56	8.31	(7)
(8)		HDD50	120	0	0	0	0	6	38	48	25	2	1	0	0	(8)
(9)		HDD65	1947	1	3	23	96	271	411	451	370	197	88	29	7	(9)
(10)		CDD50	5515	930	792	676	427	207	78	62	122	288	484	639	810	(10)
(11)		CDD65	1867	466	375	234	73	7	1	0	3	33	107	217	351	(11)
(12)		CDH74	24692	6555	4925	2765	764	56	2	1	36	427	1421	2881	4859	(12)
(13)		CDH80	12955	3878	2726	1279	230	1	0	0	4	121	581	1458	2677	(13)
(14)	Wind	WSAvg	10.7	12.2	11.9	10.8	9.4	8.9	9.1	9.6	10.2	11.0	11.7	11.9	12.0	(14)
(15)	Precipitation	PrecAvg	10.5	1.3	0.6	0.9	1.0	1.0	0.6	0.7	0.8	1.0	1.1	0.7	1.0	(15)
(16)		PrecMax	28.1	6.5	3.2	9.2	8.5	3.3	2.4	2.0	2.9	5.4	4.3	2.6	4.2	(16)
(17)		PrecMin	2.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	4.9	1.5	0.7	1.7	1.6	1.0	0.6	0.5	0.7	1.3	1.2	0.6	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	109.1	106.8	99.9	90.8	79.0	72.0	71.0	78.3	88.2	95.2	103.6	105.8	(19)
(20)			MCWB	67.0	68.6	64.9	61.5	57.4	55.5	53.3	54.9	58.5	60.9	64.5	65.2	(20)
(21)		2%	DB	104.9	101.8	95.5	85.9	75.3	67.6	66.5	72.9	83.6	91.0	97.7	101.4	(21)
(22)			MCWB	66.1	66.2	64.3	60.3	56.4	53.9	51.4	53.1	56.7	59.8	62.8	64.2	(22)
(23)		5%	DB	101.0	97.6	91.5	82.1	71.4	63.8	63.2	68.9	79.0	86.7	92.9	97.5	(23)
(24)			MCWB	66.0	65.6	63.3	58.7	55.1	52.0	50.2	51.3	55.3	58.4	61.4	63.2	(24)
(25)		10%	DB	97.0	93.9	87.3	78.2	67.8	60.9	60.4	65.3	74.4	82.1	88.6	93.4	(25)
(26)			MCWB	65.2	64.9	62.5	58.0	54.1	51.1	49.1	50.2	53.6	56.8	60.3	62.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.9	75.5	71.4	66.1	61.6	59.1	58.3	62.2	65.0	69.4	72.6	(27)	
(28)			MCD B	84.9	84.6	82.9	77.6	69.4	66.3	64.1	70.1	75.8	80.5	83.1	83.4	(28)
(29)		2%	WB	71.7	73.1	69.1	63.6	59.4	56.3	53.7	55.3	59.2	62.8	67.0	68.9	(29)
(30)			MCD B	85.5	84.3	82.0	75.1	68.6	63.7	60.5	68.1	76.3	79.5	82.3	85.0	(30)
(31)		5%	WB	69.8	71.0	66.9	61.6	57.4	54.0	51.8	52.9	57.1	60.6	65.0	66.8	(31)
(32)			MCD B	87.6	85.5	82.9	75.6	67.4	61.4	60.1	66.1	74.4	81.0	84.3	86.7	(32)
(33)		10%	WB	67.9	68.6	64.7	59.6	55.6	52.0	50.2	51.2	55.2	58.8	63.1	64.9	(33)
(34)			MCD B	89.4	86.3	82.3	75.0	65.8	59.0	58.4	62.8	71.1	78.0	83.1	88.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	24.2	22.8	22.6	21.9	19.0	17.1	17.7	20.7	22.7	23.7	23.8	24.4	(35)
(36)			MCDBR	27.4	25.8	25.9	25.7	22.8	20.2	21.3	25.0	27.8	28.8	27.9	27.3	(36)
(37)			MCWBR	10.2	9.6	10.4	10.9	10.7	10.6	11.1	11.8	11.6	11.9	10.8	10.2	(37)
(38)		5% WB	MCD BR	22.1	19.0	20.5	20.4	18.0	16.7	17.6	21.6	23.4	23.3	23.0	22.4	(38)
(39)			MCWBR	9.7	7.7	9.4	9.6	9.5	9.8	10.3	11.2	11.0	10.8	9.4	9.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.329	0.328	0.311	0.300	0.287	0.279	0.281	0.291	0.328	0.325	0.336	0.326	(40)	
(41)		taud	2.507	2.535	2.569	2.571	2.582	2.592	2.574	2.527	2.411	2.441	2.448	2.488	(41)	
(42)		Ebn at Noon	320	314	310	297	285	280	284	295	298	312	316	322	(42)	
(43)		Edh at Noon	36	34	31	27	24	22	24	28	35	37	38	37	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2574	2325	1974	1509	1107	921	1014	1333	1781	2180	2390	2583	(44)	
(45)		RadStd	143	123	128	74	61	50	47	70	102	108	130	97	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page